

TitleTrack

Pick your city job title, see what AI actually means for your work task-by-task — no fake risk scores — and see the exam that's your next move up.

Ryan Michie

Built on NYC's own payroll, exam, and occupation data

“How do we make sure AI lifts workers instead of leaving them behind?”

— the question this event asked

For 550,219 people on New York City’s payroll, that question is currently answered by headlines and generic “40% of jobs at risk” scores. Those scores are wrong, because they treat a job title as if it were one thing. The city’s own data proves it isn’t:

- **CARETAKER** is a NYCHA janitor, not a caregiver.
- **TEACHER- PER SESSION** — 78,618 people, the largest title in the city — isn’t a job at all. It’s a pay code.
- **LIEUTENANT** is two different jobs, NYPD or FDNY, mapping to two different occupation codes.
- **And nothing connects.** Payroll, exams, and occupations share no key. The join has to be rebuilt from text.

Answer it one task at a time. Never score the job.

A worker picks their real payroll title and sees the real tasks that job involves — published federal task statements, shown verbatim. Every statement gets one of three labels:

AUTOMATABLE

AUGMENTABLE

HUMAN ANCHORED

- **No overall risk score — on purpose.** Holding method constant and changing only the unit of analysis moves US “high-risk” employment from 38% to 9% (Arntz et al. 2017). The scary number is an artifact of aggregation.
- **Two kinds of human, and they are opposites.** A task can stay human because it needs a body present — which erodes as technology improves — or because the law vests the act in a person, which no model release changes. A technology roadmap and a policy roadmap. One score erases the difference.
- **It knows when to refuse.** Ask it about CITY SEASONAL AIDE and it declines: a hiring category, 17 salaried rows out of 2,414. A gated eval fails the build if that refusal is ever replaced with a guess.
- **Then it gives them a move.** Open DCAS exams for their title, plus their real promotional next rung — the actual civil-service mechanism for moving up.

Evals before features. Agents for the judgment calls.

- **Wrote the gate first, and made it fail.** A hard-fail eval re-derives every headcount, salary, exam, occupation code, and task statement from committed city snapshots. It exits non-zero on a forced match, an invented number, or any job-level aggregate. It existed before the demo did.
- **Agent 1 — title-matcher.** Joins DCAS exam titles to payroll titles without forcing matches. Its own eval hard-fails a forced match, because the naive baseline produced 12 corrupt ones.
- **Agent 2 — crosswalk-decider.** Maps a payroll title to an occupation code from repo evidence, golden-tested against 13 hand-decided titles. It returns BLOCKED rather than guessing, and it overruled my own assumptions twice.
- **A two-axis rubric, never averaged.** Can a model produce this output? And can that output take effect without a named human accountable for it? The second is law and process, not capability — so it is a ceiling, not a factor.
- **Claude on the page, deterministic code underneath.** Python does all the data work; Claude only explains it in plain language, grounded in the verified page — with a precomputed fallback so the demo never depends on the network.

AI lifting workers, in the four ways the brief asked for.

- **Navigate what AI means for their jobs.** Not a verdict — a task-level, title-specific answer for the frontline workers generic career tools ignore: traffic and school safety agents, EMTs, NYCHA caretakers, sergeants.
- **Find new opportunities.** Civil service already has the ladder. The gap is that it is pull, not push: DCAS sends no notice when you become eligible, filing windows are three weeks, and exam-to-hire runs 14–15 months. Wiring the exam feed to a notification closes that.
- **Reskill and upskill where it actually pays.** Target the augmentable tasks. The authority-anchored ones are a policy question, and no training changes them.
- **Augment the work people already do.** The result is not the intuitive one: the “low-skill” titles are the least exposed in this dataset, and the office-shaped edges are the exposed part.
- **Next: validate with the people in the titles.** Finish the remaining 23 crosswalk decisions, test the labels with DC 37 and Local 237 members, and pilot the eligibility notice with DCAS.